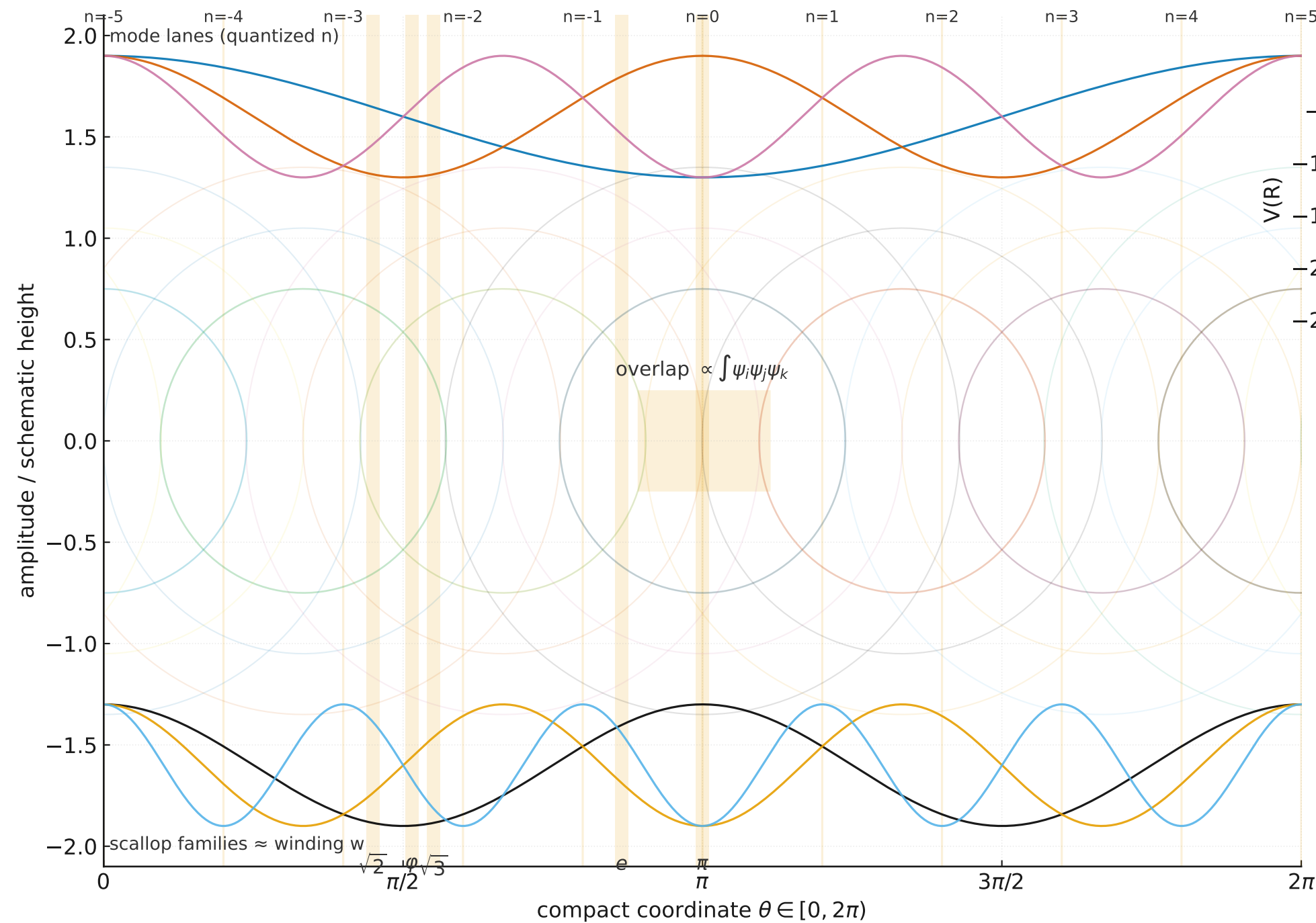
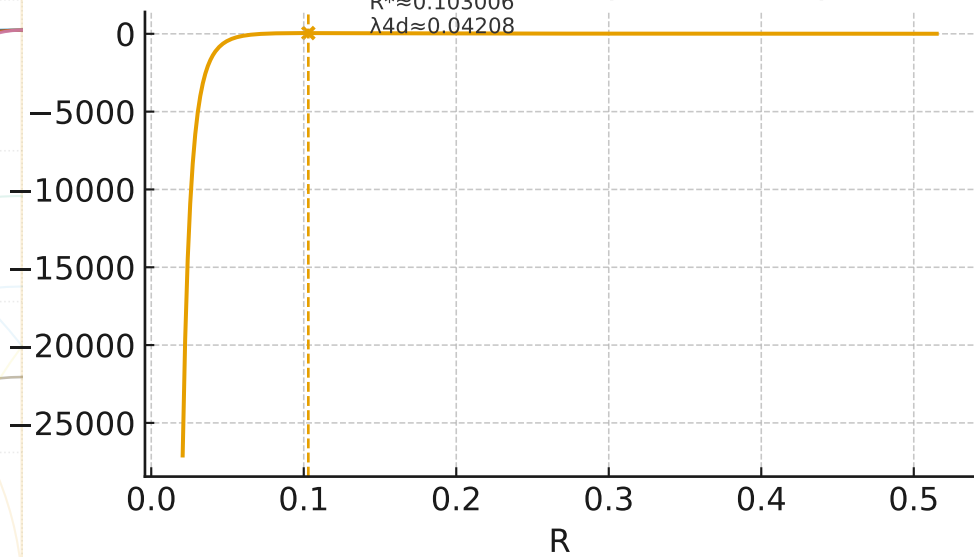


Compactification Framework: Modes, Winding, Special Loci, and Stabilized Radius



Stabilization:  $V(R) = A(1/R^2 + R^2) + B/R^4$



Numerical Anchors:

$\alpha^{-1} \approx \varphi^4 e^3 = 137.668$  (0.46%)

$m_p/m_e \approx 6\pi^5 = 1836.118$  ( $1.8 \times 10^{-5}$ )

$m_e/M_p \approx \varphi^{19} \pi^{-53} = 4.186 \times 10^{-23}$  ( $1.8 \times 10^{-4}$ )

Hypothesis: Proton/electron ratio

$m_p/m_e \approx 6\pi^5$

'6' = six compact dimensions

' $\pi^5$ ' = five winding directions

→ ratio = multiplicity × geometric factor.

Quantization on  $S^1$ :  $\psi_n(\theta) = e^{in\theta}$ ,  $k_n = \frac{n}{R}$ , spacing  $\Delta k = \frac{1}{R}$ .

KK + winding (closed strings):  $m^2 = m_0^2 + (n/R)^2 + (wR/\alpha')^2$ ,  $n, w \in \mathbb{Z}$ .

Yukawa/overlap:  $y_{ijk} \propto \int_0^{2\pi} d\theta \psi_{n_i} \psi_{n_j} \psi_{n_k}$ , selection rule  $n_i + n_j + n_k = 0$  (or mod N).

Charge from cycles: winding  $w \leftrightarrow$  charge unit. Modulus  $R$  fixes lane spacing.

5D→4D coupling:  $\lambda_{4d} = \lambda_{5d}/(2\pi R)$ . Using  $\lambda_{5d} = 0.027233$  and  $R = 1/(6\varphi) \rightarrow \lambda_{4d} \approx 0.04208$ .